

Question Number	Answer	Additional guidance	Mark
7(a)	• correct calculation (1) • correct answer to the nearest whole number (1)	<u>Example of calculation:</u> $52\,892 \times (1.016)^3 = 55471.65$ $= 55472$ alternative method $52\,892 \times (1.016) = 53\,738$ and repeat 2 more times Correct answer with no working shown scores full marks	(2)

Question Number	Answer	Mark
7(b)(i)	The only correct answer is C two <i>A is not correct because two varieties had a DI higher than 50</i> <i>B is not correct because two varieties had a DI higher than 50</i> <i>D is not correct because two varieties had a DI higher than 50</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that includes four of the following points: • seeds would be (prepared by being) dried (1) • seeds would be treated with an antimicrobial (1) • seeds would be stored in suitable conditions (1) • samples of seeds would be {germinated / x-rayed / scanned} (to check viability) (1) • replacement seeds would be collected from (these plants) (1)	accept seeds are {sterilised / disinfected} e.g. seeds are frozen, placed in temperatures below 0°C, placed in dry conditions ignore 'low temperatures' unless linked to preventing {germination / growth / microbial growth} ACCEPT (some) seeds {'allowed to grow' / 'planted'} (to check viability) ignore gather more seeds from other places	(4)

Question Number	Answer	Additional guidance	Mark
7(b)(iii)	An explanation that includes four of the following points: • set up a {breeding programme / stud book} (1) • {cross / breed} variety A with variety D (1) • (because) D has {lowest wilt percentage / lowest disease index / best resistance} and variety A has the highest yield / A and D will produce offspring with resistance and high yield (1) • infect grown offspring with the bacterium / check to see if offspring are resistant (1) • repeat using offspring with required characteristics (to get a new high-yielding and resistant variety) (1)	Accept keeping records of offspring ACCEPT analysis of more plant varieties in seed bank to see if any have more resistance to the <i>R. solanacearum</i> bacteria accept {cross/breed} high yield varieties with (disease) resistant varieties	(4)